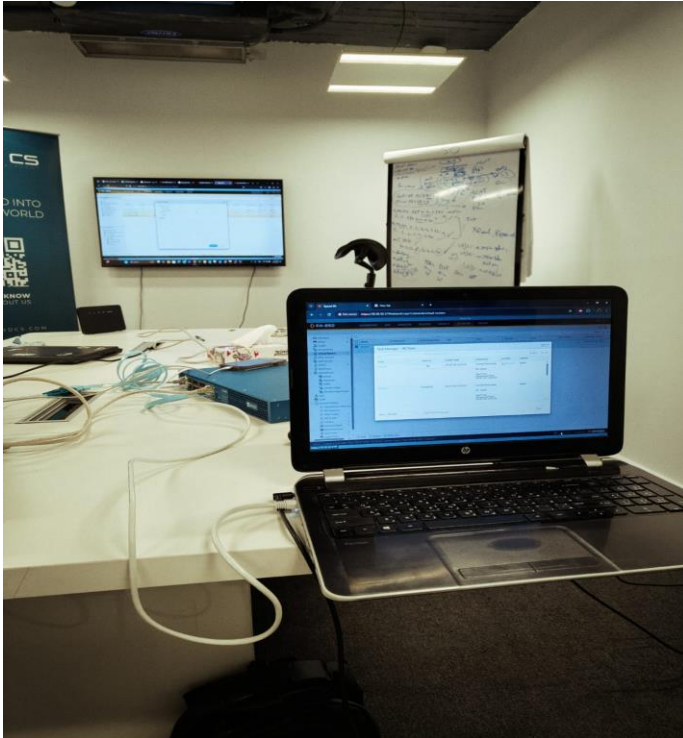




PALO ALTO LAB

ABSTRACT

Abdelrhman Hamed

Network Security Lab using Palo Alto Firewall

Objective

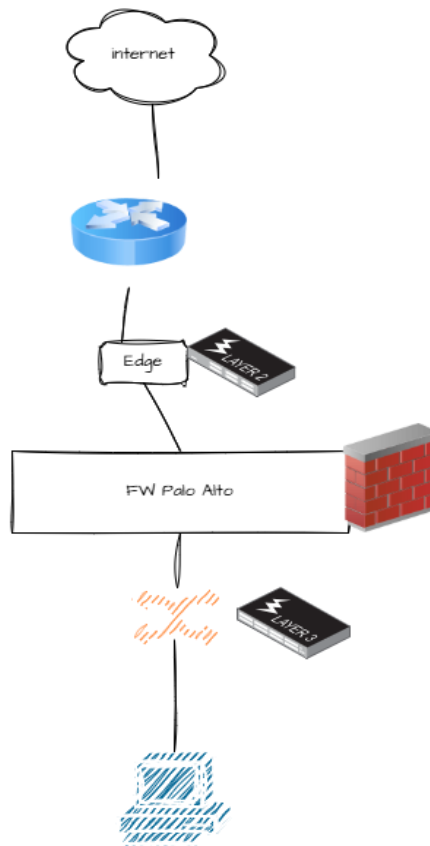
This document describes the setup and configuration of a network security lab using a Palo Alto firewall. The goal of this lab is to implement and test firewall policies, network segmentation, and secure access control.

Lab Components

The lab consists of the following devices:

1. **Internet** – External network access.
2. **Edge Router** – Connects the internal network to the internet.
3. **Palo Alto Firewall** – Provides network security and traffic control.
4. **LAN Network** – Internal users and devices.
5. **Layer 3 Switch** – Connects internal devices to the firewall.
6. **End-User System** – A computer used for testing connectivity.

Lab 1 (Palo Alto FW)



Network Topology

The network follows a structured topology where the internet connects to the edge router, which then connects to the Palo Alto firewall. The firewall manages traffic between the LAN and the external network, enforcing security policies.

Resetting the Palo Alto Firewall Password

Since the firewall settings were forgotten, the first step is to reset the administrator password.

The process involves:

1. **Reboot the firewall** and enter maintenance mode.
2. **Select "Factory Reset" or "Reset Password"** from the recovery menu.
3. **Set a new administrator password** after the reset.
4. **Reboot and access the firewall** with the new credentials.

Firewall Configuration Steps

Once access is restored, configure the firewall as follows:

1. **Interface Configuration:**
 - Assign interfaces to appropriate zones (WAN, LAN, DMZ).
 - Configure IP addresses.
2. **Security Policies:**
 - Define rules to allow/deny traffic.
 - Implement access control policies.
3. **NAT Configuration:**
 - Setup source and destination NAT for internet access.
4. **User Authentication:**
 - Configure login authentication.
 - Enable SSH and web GUI access for management.

The screenshot displays the Palo Alto PA-850 web GUI dashboard. The top navigation bar includes tabs for DASHBOARD, ACC, MONITOR, POLICIES, OBJECTS, NETWORK, and DEVICE. The main content area is divided into several sections:

- Interfaces:** A grid showing 12 interfaces (1-12) with status indicators.
- General Information:** A table listing device details for Xpand-PA, including MGT IP Address (192.168.1.3), MGT Netmask (255.255.255.0), MGT Default Gateway (192.168.1.1), MGT IPv6 Address (unknown), MGT IPv6 Link Local Address (fe80::d61d:71ff:fe91:1a00/64), MGT IPv6 Default Gateway, MGT MAC Address (d4:1d:71:91:1a:00), Model (PA-850), Serial # (011901000543), Software Version (10.2.10-h9), GlobalProtect Agent (0.0.0), Application Version (8614-7547), Device Dictionary Version (82-406), URL Filtering Version (0000.00.00.000), GlobalProtect Clientless VPN Version (0), Time (Thu Mar 27 14:07:16 2025), Uptime (0 days, 3:06:22), Advanced Routing (off), Plugin DLP (dlp-3.0.6), and Device Certificate Status (None).
- Logged In Admins:** A table showing active sessions for Admin, From, Client, Session Start, and Idle For.
- Data Logs:** A section indicating "No data available."
- System Logs:** A table listing system events, including configuration load phases and ARP conflicts.

The bottom status bar shows the user "Abdelrhman" is logged in, with a last login time of 03/27/2025 13:54:13 and a session expire time of 04/26/2025 14:03:25. A message indicates the role has been changed since the last login.

Palo Alto Firewall GUI Configuration (Device > Setup) :

In the Palo Alto firewall's web interface, I performed several configuration changes to customize the device settings:

1. Hostname and Domain Configuration

- Changed the **hostname** to `Xpand-PA` to ensure easy identification within the network.
- Set the **domain** to `mcs.local`, which is useful for internal network organization and management.

2. Login Banner Customization

- Configured a **login banner message** to display `Welcome`. This banner appears when users attempt to access the firewall interface, serving as an informative or security notice.

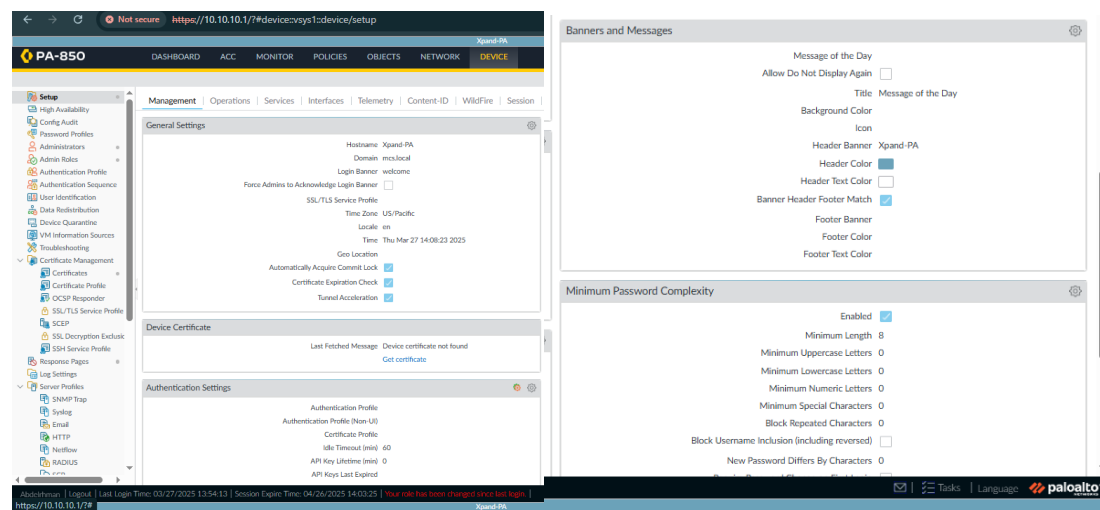
3. Time Zone Adjustment

- Updated the **time zone settings** to align with the correct geographical region. This is essential for accurate system logs, event tracking, and synchronization with other network devices.

4. Banners and Messages Customization

- Edited the **Header Banner** to `Xpand-PA`, ensuring a branded or recognizable identifier in the interface.
- Adjusted the **Header Color** and **Header Text Color** for better readability and aesthetic alignment with organizational preferences.

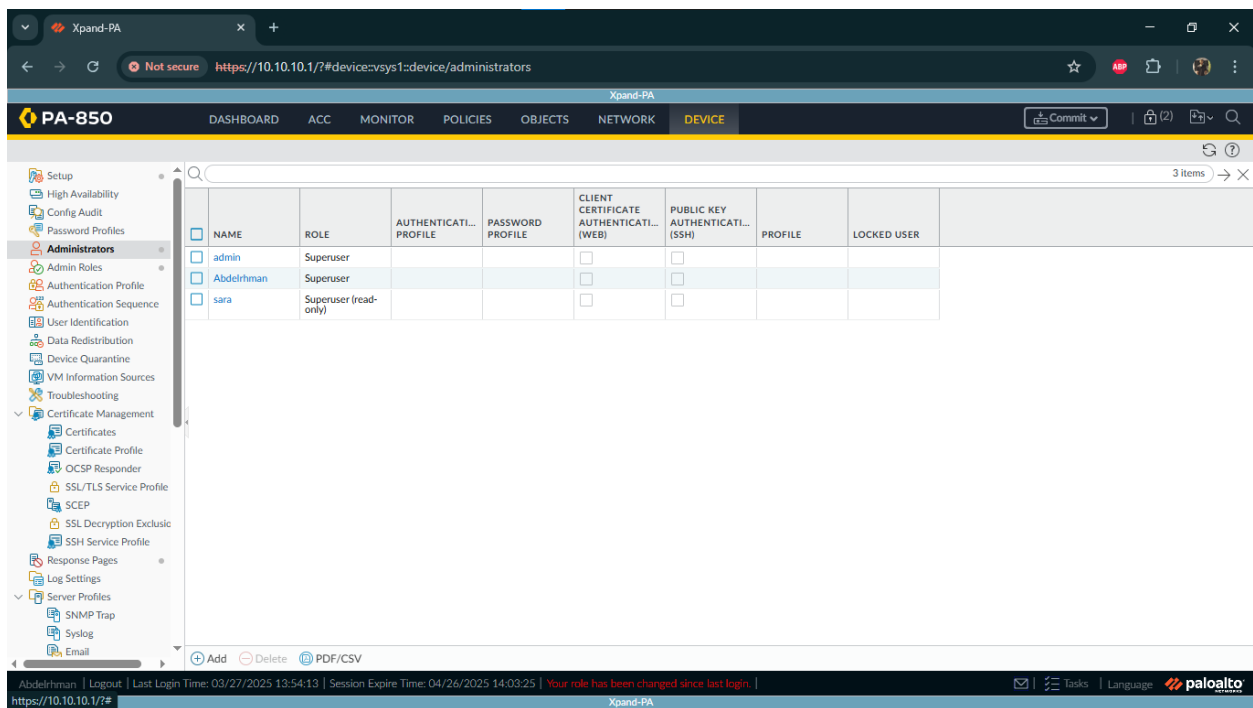
These changes enhance the firewall's usability, security, and visibility within the network environment, making it easier for administrators to manage and monitor.



GUI Configuration (Device > Administrators):

- I am created two user accounts in the Palo Alto firewall:
 1. **abdelrhman** – Assigned **Superuser** privileges (full administrative access).
 2. **sara** – Assigned **Superuser (Read-Only)** privileges (can view configurations but cannot modify them).

This setup ensures that **abdelrhman** has full control over the firewall, while **sara** can monitor configurations without making changes.



Breakdown of the Virtual Router Configuration :

1. Virtual Router Name

- **"default"** → The system comes with a default virtual router, which is used unless additional ones are manually created.

2. Interfaces Assigned

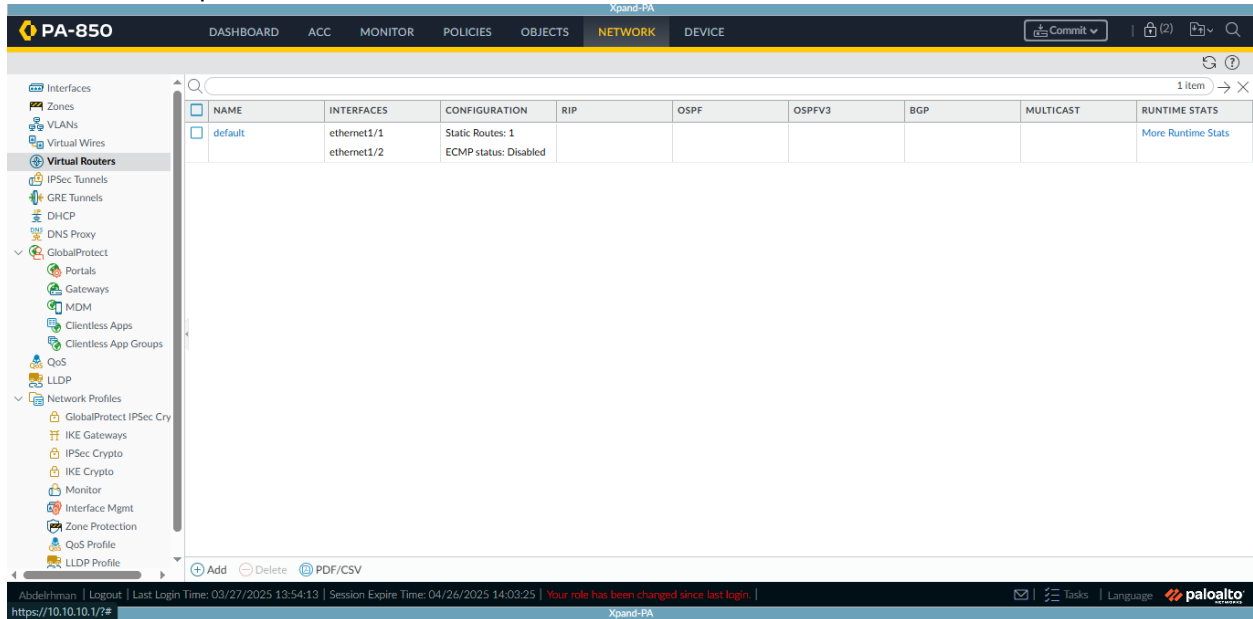
- **ethernet1/1**
- **ethernet1/2**

These interfaces are part of this virtual router, meaning they can communicate based on the routing rules defined.

3. Configuration Section

- **Static Routes: 1**
 - This means there is one manually configured static route.
 - Static routes are manually entered and do not change dynamically.

- The existing route could be used to forward traffic from one network to another.
- **ECMP (Equal-Cost Multi-Path) Status: Disabled**
 - ECMP allows load balancing across multiple routes that have the same cost.
 - Since **ECMP is disabled**, the firewall will **only use a single best route**, even if multiple paths exist.



Explanation of the NAT Policy in Palo Alto Networks

1. Understanding NAT Rule "Nating"

A single NAT rule named "Nating" is configured. Let's break down the important fields:

◆ Original Packet (Before Translation)

Field	Value	Meaning
Source Zone	DC	The traffic originates from the DC (Data Center) zone.
Destination Zone	internet	The traffic is destined for the internet .
Destination Interface	any	No specific outgoing interface is defined.
Source Address	any	The rule applies to all source IP addresses .
Destination Address	any	The rule applies to any destination IP address .
Service	any	The rule applies to all services (ports and protocols) .

Translated Packet (After NAT)

Field	Value	Meaning
Source Translation	dynamic-ip-and-port	Source NAT is performed using a Dynamic IP and Port (PAT) , meaning multiple internal devices share a single public IP but use different ports.
Pool Name	p2p_router4g	This NAT rule is using the IP pool named p2p_router4g for outbound translation.
Destination Translation	none	No destination NAT is applied (this is only Source NAT).

2. What This NAT Rule Does

- **Source NAT (SNAT):**
 - Translates private IP addresses from the **DC zone** to a **public IP from the p2p_router4g pool** before sending traffic to the internet.
 - **Dynamic IP and Port (PAT)** allows multiple internal devices to share a single external IP.
- **No Destination NAT (DNAT):**
 - This means inbound traffic **from the internet** is not being translated to an internal server.

3. Possible Use Case

This rule is typically used for **allowing internal users (in the DC zone) to access the internet** while hiding their private IP addresses behind a public IP.

The screenshot displays the Palo Alto Networks Xpand-PA interface. The top navigation bar includes 'PA-850', 'DASHBOARD', 'ACC', 'MONITOR', 'POLICIES', 'OBJECTS', 'NETWORK', and 'DEVICE'. The 'POLICIES' tab is active, showing a list of policies. The 'NAT' policy is selected, and its configuration is displayed in a table.

			Original Packet					Translated Packet			
NAME	TAGS		SOURCE ZONE	DESTINATION ZONE	DESTINATION INTERFACE	SOURCE ADDRESS	DESTINATION ADDRESS	SERVICE	SOURCE TRANSLATION	DESTINATION TRANSLATION	HIT
1	Natng	none	DC	Internet	any	any	any	any	dynamic-ip-and-port p2p_router4g	none	-

The interface also includes a 'Policy Optimizer' section on the left, showing 'Rule Usage' with 'Unused in 30 days' and 'Unused in 90 days' indicators. The bottom status bar shows the user 'Abdolkhan', login time, session expiration, and a warning: 'Your role has been changed since last login'.

The Security Policies page of a Palo Alto Networks PA-850 firewall.

1. Policies Overview

- This page defines the security policies that control traffic flow through the firewall.
- Security rules specify which traffic is **allowed or denied** based on conditions such as **source, destination, application, and service**.

2. Security Rules Breakdown

You have four security rules:

#	Name	Type	Source Zone	Destination Zone	Application
1	Dc2_block_youtube	Universal	DC	Internet	Google-base, YouTube
2	Dc2_internet	Universal	DC	Internet	Any
3	intrazone-default	Intrazone	Any	(Intrazone)	Any
4	interzone-default	Interzone	Any	Any	Any

Rule 1: Dc2_block_youtube

- **Purpose:** Blocks access to YouTube and Google-based applications from the **DC zone** to the **internet**.
- **Effect:** Any traffic originating from **DC** towards the **internet** with applications "google-base" and "youtube" will be blocked.

Rule 2: Dc2_internet

- **Purpose:** Allows general internet access for devices in the **DC** zone.
- **Effect:** Any traffic from **DC** to the **internet** is allowed without restrictions.

Rule 3: Intrazone-Default

- **Type:** Intrazone (traffic within the same zone).
- **Purpose:** Allows traffic **within the same zone** (e.g., between two devices in the same VLAN or network segment).
- **Effect:** All traffic inside the same security zone is **allowed** by default.

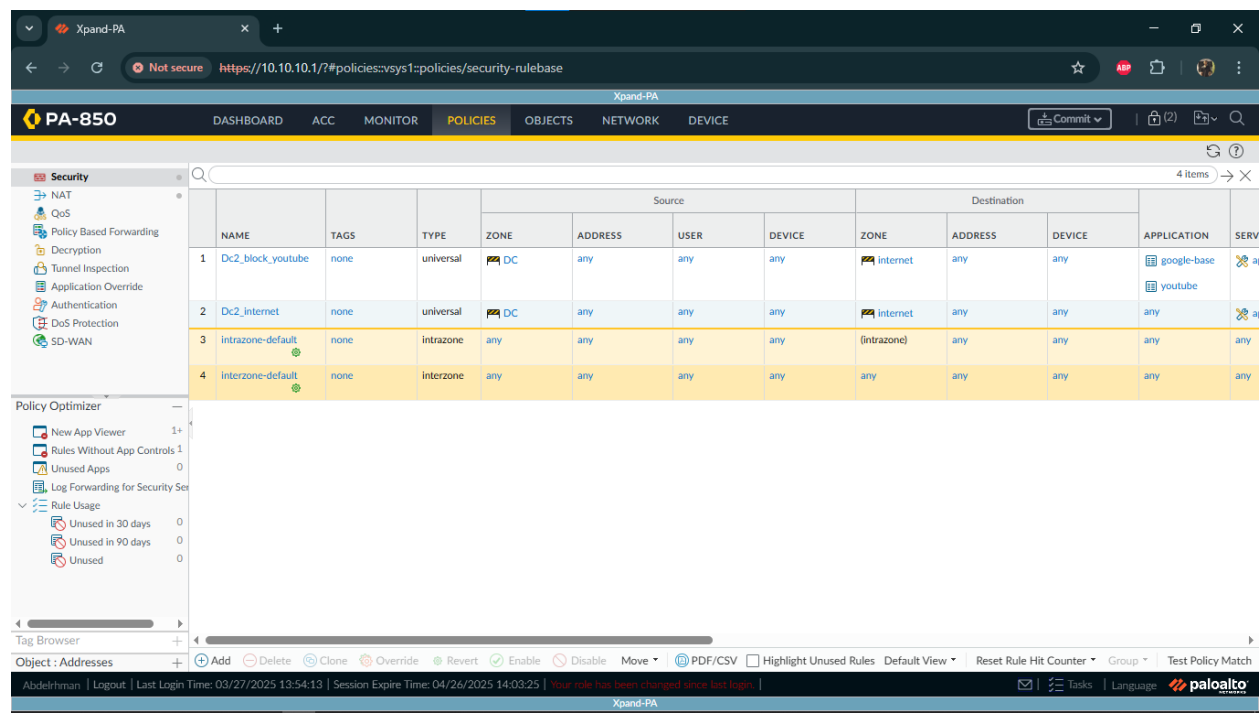
Rule 4: Interzone-Default

- **Type:** Interzone (traffic between different zones).
- **Purpose:** Defines default behavior for traffic between different zones.
- **Effect:** Traffic **between different zones** is **denied by default** unless explicitly allowed by other policies.

3. Key Observations

- The "Dc2_block_youtube" rule is specific to blocking YouTube and Google Base apps.

- The "Dc2_internet" rule allows general internet access for DC users.
- The default intrazone and interzone rules exist as fallback mechanisms.



Updated NAT Rule Analysis with Hit Count

1. Key Observations

Field	Value	Meaning
Source Address	any	The rule applies to all source IPs.
Destination Address	any	The rule applies to all destination IPs.
Service	any	The rule applies to all services (ports/protocols).
Source Translation	dynamic-ip-and-port	Source NAT (PAT) is used to translate private IPs to a public IP with port allocation.
NAT Pool	p2p_router4g	The public IP pool used for NAT.

Field	Value	Meaning
Destination Translation	none	No destination NAT is configured.
Hit Count	9002	This NAT rule has been used 9002 times since creation.
Last Hit	2025-03-27 14:10:10	The most recent time traffic matched this rule.
First Hit	2025-03-27 13:25:57	The first recorded usage of this rule.
Modified	2025-03-27 13:23:27	Last modification time.
Created	2025-03-27 13:23:27	The rule was created at this time.

The screenshot shows the Palo Alto Networks PA-850 web interface. The top navigation bar includes 'DASHBOARD', 'ACC', 'MONITOR', 'POLICIES', 'OBJECTS', 'NETWORK', and 'DEVICE'. The 'POLICIES' tab is selected, and the 'NAT' rule is highlighted in the left sidebar. The main content area displays a table with the following columns: SOURCE ADDRESS, DESTINATION ADDRESS, SERVICE, SOURCE TRANSLATION, DESTINATION TRANSLATION, HIT COUNT, LAST HIT, FIRST HIT, MODIFIED, and CREATED. The table contains one row with the following values: any, any, any, dynamic-ip-and-port, none, 9002, 2025-03-27 14:10:10, 2025-03-27 13:25:57, 2025-03-27 13:23:27, and 2025-03-27 13:23:27. The bottom status bar shows the user 'Abdulkhman' and the session expiration time '04/26/2025 14:03:25'.

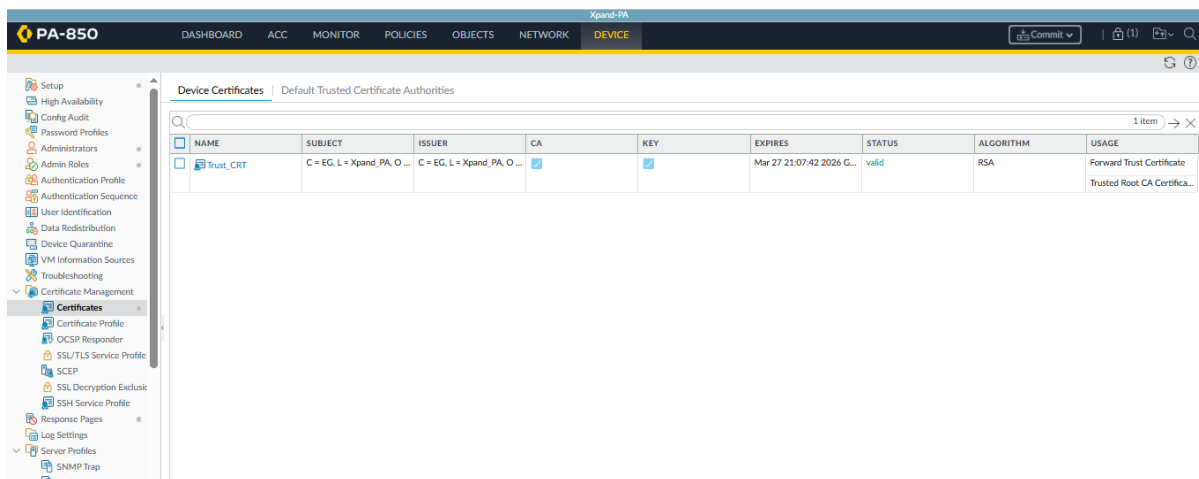
SOURCE ADDRESS	DESTINATION ADDRESS	SERVICE	SOURCE TRANSLATION	DESTINATION TRANSLATION	HIT COUNT	LAST HIT	FIRST HIT	MODIFIED	CREATED
any	any	any	dynamic-ip-and-port	none	9002	2025-03-27 14:10:10	2025-03-27 13:25:57	2025-03-27 13:23:27	2025-03-27 13:23:27

Detailed Explanation of Certificate Details:

1. Subject & Issuer:

- The certificate was **issued to** and **signed by** "Xpand, PA," meaning it is a **self-signed certificate**.
- Self-signed certificates are created and signed by the same entity rather than a publicly trusted Certificate Authority (CA) like DigiCert or GlobalSign.

- Used for internal security purposes within the firewall.
- 2. **CA (Certificate Authority):**
 - This certificate is marked as a **Certificate Authority (CA)**, meaning it can issue and sign other certificates.
 - Typically, firewalls like Palo Alto use internal CA certificates for SSL/TLS decryption (e.g., decrypting HTTPS traffic for inspection).
- 3. **Key:**
 - The **private key** associated with this certificate is present, enabling encryption and decryption.
 - If this key were missing, the certificate wouldn't be functional for encryption-related tasks.
- 4. **Expiration Date:**
 - The certificate is valid until **March 27, 2026, at 21:07:42 GMT**.
 - After this date, the certificate will expire, and firewall services relying on it (like SSL decryption) may stop working unless renewed.
- 5. **Status:**
 - The certificate is **valid**, meaning it has not expired or been revoked.
 - If the status were "expired" or "revoked," it would cause security warnings and failures in SSL decryption.
- 6. **Algorithm:**
 - Uses the **RSA (Rivest-Shamir-Adleman)** encryption algorithm.
 - RSA is a widely used asymmetric encryption method for securing network communications.
- 7. **Usage:**
 - **Forward Trust Certificate:**
 - This means the firewall uses this certificate to sign and decrypt SSL traffic when performing **SSL/TLS inspection** (e.g., Man-in-the-Middle decryption for security policies).
 - If a client trusts this certificate, they will accept connections without SSL errors.
 - **Trusted Root CA Certificate:**
 - The firewall acts as a **trusted root CA**, meaning devices in the network can trust it for issuing and validating certificates.
 - It is essential for SSL decryption and preventing security warnings in browsers.



NAME	SUBJECT	ISSUER	CA	KEY	EXPIRES	STATUS	ALGORITHM	USAGE
Trust_CRT	C = EG, L = Xpand, PA, O ...	C = EG, L = Xpand, PA, O ...	☒	☒	Mar 27 21:07:42 2026 G...	valid	RSA	Forward Trust Certificate Trusted Root CA Certifica...

Lab Summary: Palo Alto Firewall Deployment and Network Security Configuration

This lab focuses on setting up and configuring a **Palo Alto Networks (PA-850) firewall** in a network security environment. The objective is to deploy, manage, and troubleshoot security policies, routing, NAT, and certificates within a structured network topology.

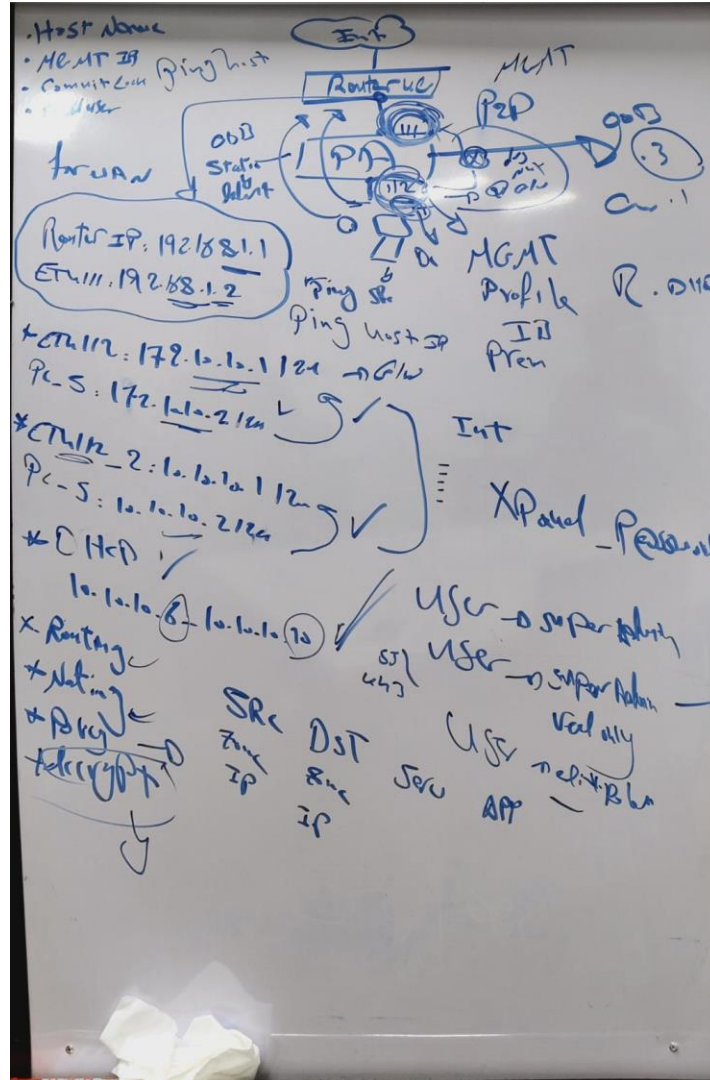
Network Topology Overview

1. **Internet Edge Router**
 - Connects the network to the internet.
 - Provides external connectivity and interacts with the firewall.
2. **Edge Switch**
 - Acts as an intermediary between the firewall and the core network.
 - Ensures proper VLAN configuration and forwarding rules.
3. **Palo Alto Firewall**
 - Core component for network security.
 - Handles firewall policies, NAT, routing, and user authentication.
 - Provides security filtering, logging, and traffic control.
4. **Internal Network & Hosts**
 - Includes internal workstations and servers.
 - Users authenticate and interact with resources while following security policies.

Key Configurations and Findings

- **Firewall Certificate Management**
 - A self-signed **Root CA certificate** ("Trust_CRT") is configured and used for SSL decryption and trusted CA functions.
 - The certificate is marked as valid, using the **RSA algorithm**, and expires in **2026**.
 - This certificate is used for **Forward Trust Certificate** and **Trusted Root CA Certificate** functionalities.
- **IP Addressing & Connectivity Testing**
 - The **router** is configured with **192.168.1.1**, and a connected **host** with **192.168.1.2**.
 - Multiple VLANs and subnets are assigned for different segments, such as **172.16.1.x** and **10.1.1.x**.
 - **Ping tests** confirm connectivity and troubleshooting steps for network reachability.
- **Routing, NAT, and Security Policies**
 - **Routing:** Static and dynamic routing is configured to manage traffic flow between internal and external networks.
 - **NAT:** Network Address Translation (NAT) is implemented to map internal IPs to public-facing IPs.
 - **Security Policies:** Rules are applied to **filter traffic**, enforce access control, and inspect packets.
- **User Authentication & Role-Based Access**
 - Super Admin and Admin roles are defined for managing the firewall.
 - Policies enforce **user authentication** before accessing resources.

- Logging and monitoring features track user activity.



Conclusion

This lab provides hands-on experience with **Palo Alto firewall administration**, covering:

- Firewall policy creation
- Certificate management for secure connections
- Routing, NAT, and VLAN configurations
- User authentication and security enforcement

Thank for all

